

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 04/08/2020

Quiz 9: Section 4.1, 4.2, 4.3

Name (Print): _____

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Section 4.1

3 pts

1. Find the absolute max/min **values** of $f(x) = x^4 - 2x^3$ on the interval $[-2, 2]$.

- (A) The absolute max. of $f(x)$ on $[-2, 2]$ is 32 and the absolute min. is 0.
- (B) The absolute max. of $f(x)$ on $[-2, 2]$ is $3/2$ and the absolute min. is $-27/16$.
- (C) **The absolute max. of $f(x)$ on $[-2, 2]$ is 32 and the absolute min. is $-27/16$.**
- (D) The absolute max. of $f(x)$ on $[-2, 2]$ is $3/2$ and the absolute min. is -2 .

Section 4.2

2 pts

1. Find an interval $I = [a, b]$ on which Rolle's Theorem applies to $f(x) = x^3 - 4x^2 + 4x$. Then find the point(s) c in (a, b) at which $f'(c) = 0$.

- Ⓐ $I = [0, 2]$, $c = 2$ and $c = \frac{2}{3}$
- Ⓑ $I = [0, 2]$, $c = \frac{2}{3}$
- Ⓒ $I = [-2, 2]$, $c = -2$ and $c = 2$
- Ⓓ $I = [-2, 2]$, $c = \frac{2}{3}$

Section 4.3

2 pts

1. Use the Second Derivative Test to locate the local extrema of the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$ on $(-2, 2)$.
- Ⓐ $f(x)$ has a local minimum at $x = -1$ and a local maximum at $x = 1$.
 - Ⓑ $f(x)$ **has a local minimum at $x = -1$ and no local maximum.**
 - Ⓒ $f(x)$ has no local extrema on $(-2, 2)$.
 - Ⓓ $f(x)$ has a local minimum at $x = 2$ and a local maximum at $x = -2$.

3 pts

2. Identify the intervals on which the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$ is concave up or concave down. Then, locate the inflection points.

- Ⓐ $f(x)$ is concave up on $(1, \infty)$, concave down on $(-\infty, 1)$, and has an inflection point at $x = 1$.
- Ⓑ **$f(x)$ is concave up on $(-\infty, -1/3) \cup (1, \infty)$, concave down on $(-1/3, 1)$, and has inflection points at $x = -1/3$ and $x = 1$.**
- Ⓒ $f(x)$ is concave up on $(-\infty, -1/3)$, concave down on $(-1/3, \infty)$, and has an inflection point at $x = -1/3$.
- Ⓓ $f(x)$ is concave up on $(-\infty, \infty)$ and has no inflection points.